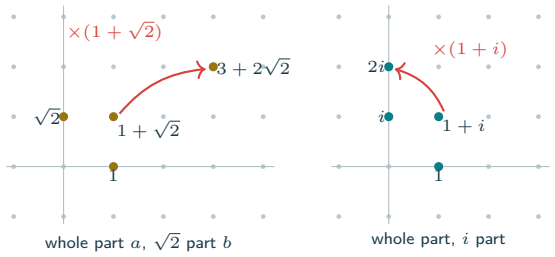
ARITHMETIC ON ONE PAGE

Adding one new number

Numbers $a + b\sqrt{2}$ can be added, subtracted and multiplied freely, and dividing takes one more idea.

1 / THE RULE AND THE PICTURE

Add a number that squares to 2, and products stay of the form $a + b\sqrt{2}$



Start with the whole numbers and one new number, $\sqrt{2}$, which satisfies $\sqrt{2} \times \sqrt{2} = 2$. Every number $a + b\sqrt{2}$ with whole a, b is a point (a, b) on a grid, sums add the coordinates, and products follow from $\sqrt{2}\sqrt{2} = 2$:

$$(a + b\sqrt{2})(c + e\sqrt{2}) = (ac + 2be) + (ae + bc)\sqrt{2}$$

For example $(1 + \sqrt{2})(3 + \sqrt{2}) = 3 + \sqrt{2} + 3\sqrt{2} + 2 = 5 + 4\sqrt{2}$, and $(1 + \sqrt{2})^2 = 3 + 2\sqrt{2}$, the arrow on the left. Adding i with $i^2 = -1$ gives the same picture, with $(1 + i)^2 = 2i$.

2 / THREE CHOICES OF NEW NUMBER

Only the sign and size of the square change the rule

new number	its square	product of two numbers of the form	twin product
$\sqrt{2}$	2	$(a + b\sqrt{2})(c + e\sqrt{2}) = (ac + 2be) + (ae + bc)\sqrt{2}$	$a^2 - 2b^2$
i	-1	$(a + bi)(c + ei) = (ac - be) + (ae + bc)\,i$	$a^2 + b^2$
$\sqrt{-2}$	-2	$(a + b\sqrt{-2})(c + e\sqrt{-2}) = (ac - 2be) + (ae + bc)\sqrt{-2}$	$a^2 + 2b^2$

The twin of $a + b\sqrt{d}$ is $a - b\sqrt{d}$, and multiplying twins always gives a whole number, the twin product $a^2 - db^2$. Later sheets call it the norm.

3 / DIVIDING: FROM A RING TO A FIELD

The twin turns division into multiplication

Because $(a + b\sqrt{2})(a - b\sqrt{2}) = a^2 - 2b^2$, we have $\dfrac{1}{a + b\sqrt{2}} = \dfrac{a - b\sqrt{2}}{a^2 - 2b^2}$, so $\frac{1}{1+\sqrt{2}} = \frac{1-\sqrt{2}}{1-2} = \sqrt{2} - 1$ stays among the whole-number numbers, but $\frac{1}{2+\sqrt{2}} = \frac{2-\sqrt{2}}{2} = 1 - \frac{1}{2}\sqrt{2}$ needs a half.

Allowing a and b to be fractions repairs this, and the denominator $a^2 - 2b^2$ is never zero unless $a = b = 0$, because $\sqrt{2}$ is irrational and $a^2 = 2b^2$ has no solution in whole numbers.

For i the denominator $a^2 + b^2$ is positive whenever a, b are not both 0.

	a, b whole numbers	a, b fractions
add, subtract, multiply	always stays inside	always stays inside
divide by a nonzero number	sometimes: $\frac{1}{1+\sqrt{2}} = \sqrt{2} - 1$ is inside, $\frac{1}{2+\sqrt{2}} = 1 - \frac{1}{2}\sqrt{2}$ is not	always stays inside
what it is called	a ring, written $\mathbb{Z}[\sqrt{2}]$	a field, written $\mathbb{Q}(\sqrt{2})$

4 / NUMBERS YOU CAN DIVIDE BY WITHOUT LEAVING THE RING

Exactly those whose twin product is 1 or -1

new number	numbers with a whole-number inverse	how many
$\sqrt{2}$	$1 + \sqrt{2},\ 3 + 2\sqrt{2},\ 7 + 5\sqrt{2},\ 17 + 12\sqrt{2},\ 41 + 29\sqrt{2}, \ldots$	infinitely many, twin products $-1, 1, -1, 1, -1, \ldots$
i	$1,\ -1,\ i,\ -i$	just four
$\sqrt{-2}$	$1,\ -1$	just two

Each $\sqrt{2}$ entry is the one before multiplied by $1 + \sqrt{2}$, and $(3 + 2\sqrt{2})(3 - 2\sqrt{2}) = 1$ shows the inverse is the twin.

5 / WHY IT MATTERS: A PRIME CAN BREAK

A prime of the whole numbers splits exactly when it is a twin product

prime	with i	with $\sqrt{-2}$	with $\sqrt{2}$
3	stays prime	$(1 + \sqrt{-2})(1 - \sqrt{-2})$	stays prime
5	$(2 + i)(2 - i)$	stays prime	stays prime
7	stays prime	stays prime	$(3 + \sqrt{2})(3 - \sqrt{2})$
17	$(4 + i)(4 - i)$	$(3 + 2\sqrt{-2})(3 - 2\sqrt{-2})$	$(5 + 2\sqrt{2})(5 - 2\sqrt{2})$

For instance $5 = 2^2 + 1^2$, $3 = 1^2 + 2 \cdot 1^2$ and $7 = 3^2 - 2 \cdot 1^2$, while 17 is a twin product in all three ways. Which new number breaks which primes is the pattern of the next sheets.

ASIDE / TRY THESE

$(2+\sqrt{2})(1+\sqrt{2}) = 4+3\sqrt{2}$; $(1+2i)(3-i) = 5+5i$; $\frac{1}{3+\sqrt{2}} = \frac{3-\sqrt{2}}{7}$ needs sevenths.

LINKS TO OTHER SHEETS

The next sheets give the general picture through the cyclotomic fields and then the pair $\sqrt{2}$, $\sqrt{-2}$.

Convention. a, b, c, e whole numbers in the ring and fractions in the field; $\mathbb{Z}[\sqrt{2}]$ is the ring and $\mathbb{Q}(\sqrt{2})$ the field; the twin product is the norm of later sheets.

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